

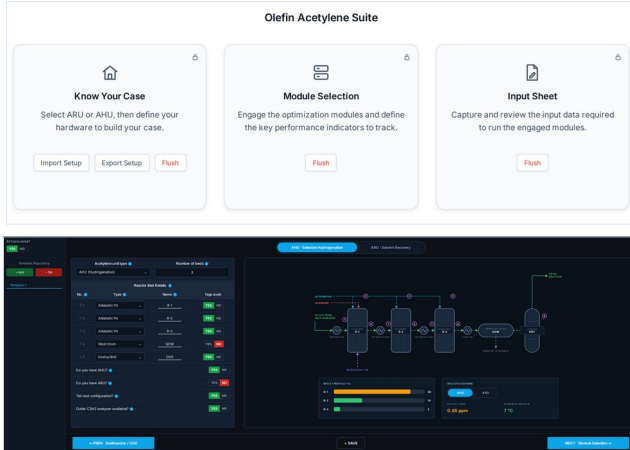
PRODUCT OVERVIEW

AcetyleneX360 is a decision-support application for acetylene removal in the olefin plant. At configuration the site selects its scheme – AHU (selective hydrogenation: front-end or tail-end, lead / guard beds, H₂ and CO dosing, green oil wash) or ARU (solvent recovery: absorber, stripper, solvent regenerator, acetylene product) – and the modules, input sheet and PFD adapt to it. Domain and ML agents track selectivity, acetylene slip, bed temperature rise and runaway margin, catalyst cycle, or for ARU, absorption, solvent losses and regeneration duty. Output reaches the process engineer as ranked, quantified actions, each reported against the parameter that caused it. Every agent is configured, validated and commissioned per site by Ingenero process engineers.



01 Unit Configuration

Shown: Olefin acetylene suite – ARU / AHU selection, configuration and PFD

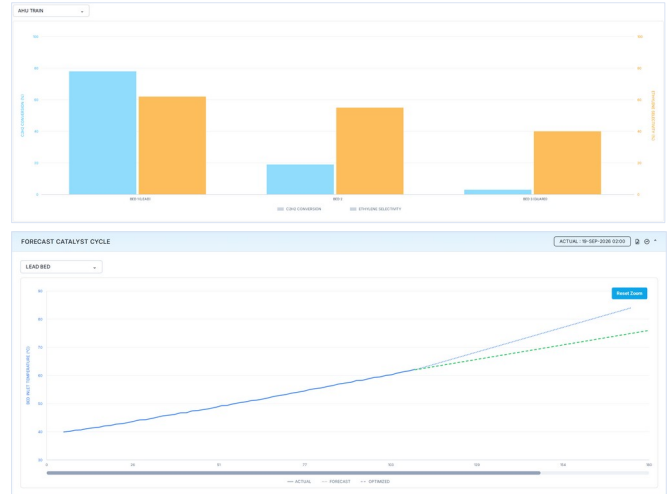


- Know Your Case** Select ARU or AHU first, then configure beds / columns, dosing and solvent
- Module Selection** Modules offered adapt to the ARU or AHU scheme – Domain & AI agents
- Input Sheet** Based on scheme & module selection – relevant sensor info needs to be filled
- Interactive Config** AHU reactor train or ARU absorber / stripper interactive PFD available



02 Plant Data and Model Execution

Shown: bed-wise conversion & selectivity, catalyst cycle forecast vs optimum.



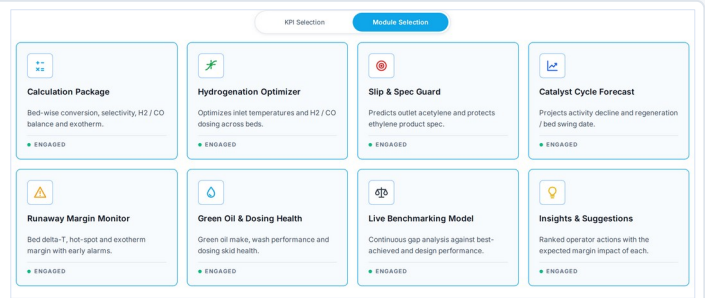
- Inputs** Sensor tags, C₂H₂ / CO / H₂ analysers and catalyst / solvent design data
- Processing** tag validation, steady state framing and bed / cycle state identification
- Outputs** predicted KPIs, ranked actionable with cause, trends, alerts



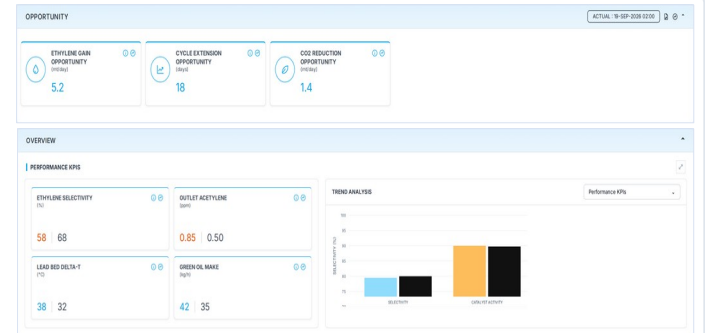
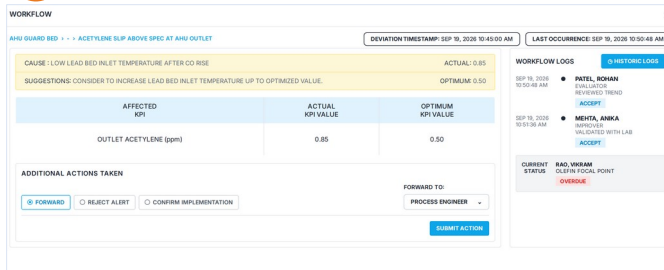
03 AI-Powered Model Agent Suite

Agents for AHU and ARU – only the set for the selected scheme is engaged

- Bed Selectivity · AHU** Ethylene gain / loss per bed with H₂ and CO moderation guidance.
- Acetylene Slip · AHU** Predicts outlet acetylene vs spec and warns before off-spec.
- Runaway Margin · AHU** Tracks bed delta-T and exotherm margin to prevent runaway.
- Catalyst Cycle · AHU** Forecasts activity, green oil build-up and regeneration date.
- Solvent Absorber · ARU** Tracks acetylene absorption, solvent rate and ethylene co-absorption.
- Solvent Regenerator · ARU** Optimizes stripper / regenerator duty, solvent purity and losses.
- Product Quality · ARU** Soft sensor for acetylene product purity and safe handling limits



04 Opportunity and Operator Actions



+20%

Catalyst cycle length

CO moderation & inlet temperature control

+0.3%

Ethylene gain

selectivity optimization across beds

5-8%

ARU solvent & energy

absorber / stripper optimization

Basis: optimization respects ethylene spec, CGC / cold section and runaway safety limits; suggestions take care of step change for smooth implementation.



05 Platform

Historian connectivity

plug and play with leading plant historians

Simulation integration

third party simulators integrate without custom work

Daily reporting

downloadable daily reports with interactive trends

Gen AI assistant

answers plant questions in the operating workflow

Build your own dashboard

engineers build the views their unit needs

DELIVERY

Configured, validated and commissioned by Ingenero process engineers.

FLEXIBILITY

ARU or AHU; front-end or tail-end AHU; selectable KPI set

SCALE

Repeat sites reuse the configuration, with no/minimal rework per plant.

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